

# Sign Errors in Parabola Transformations

## Answer Key

### Algebra 1 Quadratic Functions

#### Quick Answers

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|-------|----------|-------|----------|
| 1. 2  | 3. -3, 5 | 5. 0  | 7. -3, 7 |
| 2. -3 | 4. 7     | 6. -5 | 8. —     |
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#### Curriculum

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**Level:** Algebra 1 Quadratic Functions

HSA-APR.B.3 – *problems 3, 7*

HSA-SSE.B.3 – *problems 1, 2, 4, 5, 6, 8*

*Difficulty 1–5 of 5 across 8 tagged checks.*

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#### Common wrong answers

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1. *If they got 102: read  $x - 5$  as a vertex at  $x = -5$  instead of  $x = 5$*
  2. *If they got 61: wrote  $(x - 4)$  squared, minus 3, for a shift LEFT 4, then evaluated at  $x = -4$*
  4. *If they got -137: substituted  $x = -6$ , reading the sign inside the parentheses literally*
  5. *If they got 24: dropped the negative on the leading coefficient, so the reflection was lost*
  6. *If they got 191: read  $x + 7$  as a shift RIGHT 7 and evaluated the minimum at  $x = 7$*
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**Grading note.** On this sheet the sign reasoning is the skill, so a correct number reached by the wrong substitution should not earn full credit. The “common wrong answers” list above tells you which misreading produced each wrong value.

#### 1. Minimum value of $f(x) = (x - 5)^2 + 2$ .

The vertex  $x$  is whatever makes the squared factor zero:  $x - 5 = 0$  gives  $x = 5$ , so the vertex is at  $x = 5$ , *not*  $x = -5$ . (The graph moved right, in the opposite direction from the printed sign.)

$$f(5) = (5 - 5)^2 + 2 = 0 + 2$$

Because  $(x - 5)^2 \geq 0$  for every  $x$ , adding 2 can never produce anything smaller, so this really is the minimum. 2

*If they got 102: they substituted  $x = -5$ , reading the printed sign literally.*

**2.  $y = x^2$  shifted 4 left and 3 down; evaluate at  $x = -4$ .**

A shift left is written with a *plus* inside the parentheses, because the graph moves to where  $x + 4 = 0$ , i.e.  $x = -4$ :

$$y = (x + 4)^2 - 3$$

$$y(-4) = (-4 + 4)^2 - 3 = 0 - 3$$

The value at  $x = -4$  is the vertex height, which is the whole point of the check. -3

*If they got 61:* they wrote  $(x - 4)^2 - 3$  for “left 4” and then evaluated it at  $x = -4$ , so the two sign choices compounded.

**3.  $x$ -intercepts of  $p(x) = (x - 1)^2 - 16$ .**

Set  $p(x) = 0$  and undo the transformations in reverse order:

$$(x - 1)^2 - 16 = 0$$

$$(x - 1)^2 = 16$$

$$x - 1 = \pm 4 \quad (\text{both roots — dropping } -4 \text{ loses an intercept})$$

$$x = 1 + 4 = 5 \quad \text{or} \quad x = 1 - 4 = -3$$

Sign check: the axis of symmetry is  $x = 1$ , and the two intercepts sit 4 units on either side of it ✓

$$\boxed{x = -3 \quad \text{and} \quad x = 5}$$

**4. Find and fix: maximum of  $y = -(x - 6)^2 + 7$ .**

*The error:* Jamal read  $(x - 6)$  as a vertex at  $x = -6$ . The vertex sits where the squared factor is zero, so  $x - 6 = 0$  gives  $x = 6$ . His arithmetic was fine; his  $h$  was not.

Correct substitution:

$$y(6) = -(6 - 6)^2 + 7 = -0 + 7$$

Since  $a = -1 < 0$  the parabola opens downward, so this vertex value is a *maximum* — 7 is the largest output the function ever reaches. 7

*If they got -137:* they repeated Jamal’s substitution  $x = -6$  instead of fixing it.

**5. Reflect, stretch by 3, left 2, up 12; evaluate at  $x = 0$ .**

Build the equation one transformation at a time. Reflection and vertical stretch both live in  $a$ :  $a = -3$ . Left 2 puts +2 inside the parentheses; up 12 adds 12 outside.

$$q(x) = -3(x + 2)^2 + 12$$

$$q(0) = -3(0 + 2)^2 + 12 = -3(4) + 12 = -12 + 12$$

So  $x = 0$  is one of the  $x$ -intercepts of the transformed parabola. 0

*If they got 24:* they kept  $a = +3$  and lost the reflection. Note the vertex is at  $x = -2$  with value 12, so a positive  $a$  would make 12 a minimum instead of a maximum — the shape of the answer changes, not just the number.

**6. Find and fix: minimum of  $g(x) = (x + 7)^2 - 5$ .**

*The error:* Mia called  $(x + 7)$  a shift 7 units *right*. It is a shift 7 units *left*, because  $x + 7 = 0$  at  $x = -7$ . Her “down 5” was correct.

Because she put the vertex at  $x = 7$ , she evaluated at the wrong place. Correct substitution:

$$g(-7) = (-7 + 7)^2 - 5 = 0 - 5$$

-5

*If they got 191:* that is  $g(7)$  — Mia’s value. It is a real point on the parabola, just not the lowest one, which is exactly why a wrong  $h$  can go unnoticed.

**7. Reflect, right 2, up 25; find the  $x$ -intercepts.**

Reflection makes  $a = -1$ ; right 2 gives  $(x - 2)$ ; up 25 adds 25:

$$y = -(x - 2)^2 + 25$$

Set it to zero:

$$-(x - 2)^2 + 25 = 0$$

$$(x - 2)^2 = 25$$

$$x - 2 = \pm 5$$

$$x = 7 \quad \text{or} \quad x = -3$$

$x = -3 \quad \text{and} \quad x = 7$

*Why the reflection does not move the axis:* reflecting over the  $x$ -axis replaces each output  $y$  with  $-y$  and leaves every input untouched. The axis of symmetry is a statement about inputs, so it stays at  $x = 2$  — and indeed  $-3$  and  $7$  are each 5 units from 2.

**8. Error analysis — Ravi’s study card.** (open response — read the explanations)

**R1 is wrong.** The shift is not in the direction of the printed sign. The vertex lands where the squared factor is zero, so  $(x + 4)$  puts the vertex at  $x = -4$ : the graph moves *left* 4. *Corrected:* “In  $y = a(x - h)^2 + k$  the graph shifts to  $x = h$ , the value that makes  $x - h$  zero. So  $(x + 4)$  means  $h = -4$ : left 4.”

**R2 is correct.**  $k$  is added to every output after the squaring, so a positive  $k$  raises the whole graph and a negative  $k$  lowers it. Here the printed sign and the direction agree, because nothing is subtracted *from*  $y$ .

**R3 is wrong.**  $a$  multiplies the outputs; it does nothing to the inputs. A negative  $a$  flips the parabola upside down about its own vertex, but the vertex — and therefore both shifts — stays exactly where  $h$  and  $k$  put it. *Corrected:* “When  $a < 0$  the parabola opens downward and its vertex becomes a maximum instead of a minimum; the shifts are unchanged.”

*Full credit* requires naming R1 and R3 as wrong, an explanation that appeals to “what makes the parentheses zero” (R1) and to “ $a$  scales outputs, not inputs” (R3), and a corrected wording for each. Naming the wrong rules with no reason is partial credit.

R1 wrong, R2 correct, R3 wrong